

**Lab 6: FFT**

**Instructions:**

1. Lab report to be submitted (on Nalanda) at the end of the lab hours (preferably), late submission will result in loss of marks, use the lab hours to finish the work
  2. All the tasks in each lab have to be in one main MATLAB file with “.m” extension, differentiate each question by commenting. Alternatively, you can zip all .m files together, name them appropriately and upload one zip/rar file. Make one Word/pdf file with all the results corresponding to the questions
  3. Plots should include x label, y label, legend, and titles
  4. File name: ID\_Lab6.m/ID\_Lab6.rar or zip **and** ID\_Lab6.pdf
  5. If the report/code is found to be copied, zero marks will be awarded [**Ensure that you show the outputs of the task to the TAs and get them evaluated. No queries will be entertained post lab session.**]
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1. Explore the functions **fft** and **ifft** in MATLAB. Take sequence  $x(n)$  which we solved in the lecture, to verify the final outcome. Comment on the technique used in these functions.

$$x(n) = \{2, 1, 2, 1, 1, 2, 1, 2\}$$

In MATLAB, there is no direct command to draw the butterfly diagram, explore how can you plot a butterfly diagram.

2. Let  $x(n)$  be a finite length sequence given by

$$x(n) = \left\{ \dots, 0, 0, 0, \underset{\uparrow}{1}, 2, -3, 4, -5, 0, \dots \right\}$$

Determine and sketch the sequence  $x((-8-n))_{\mathcal{R}_7}(n)$  where

$$\mathcal{R}_7(n) = \begin{cases} 1, & 0 \leq n \leq 6 \\ 0, & \text{elsewhere} \end{cases}$$

3. The following finite-duration sequences are called windows and are very useful in DSP filter designs.

$$\text{Rectangular: } \mathcal{R}_M(n) = \begin{cases} 1, & 0 \leq n < M \\ 0, & \text{otherwise} \end{cases};$$

$$\text{Hanning: } \mathcal{C}_M(n) = 0.5 \left[ 1 - \cos \frac{2\pi n}{M-1} \right] \mathcal{R}_M(n)$$

$$\text{Triangular: } \mathcal{T}_M(n) = \left[ 1 - \frac{|M-1-2n|}{M-1} \right] \mathcal{R}_M(n);$$

$$\text{Hamming: } \mathcal{H}_M(n) = \left[ 0.54 - 0.46 \cos \frac{2\pi n}{M-1} \right] \mathcal{R}_M(n)$$

Plot the time domain signals to understand what these windows look like. Do not use direct commands for the windows, you need to create the windows based on the formula given.

For each of these windows, determine their DTFTs for  $M = 10, 25, 50, 101$ . Scale transform values so that the maximum value is equal to 1. Plot the magnitude of the normalized DTFT over  $-\pi \leq \omega \leq \pi$ . Study these plots and comment on their behavior as a function of  $M$ .

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